Voltage Protector - Project Report

1. Goal

Build a microcontroller-based device that cuts power to an appliance when the grid voltage surges or the current is too high, lets the user change the settings from the device or from a computer, and does not reconnect carelessly afterwards. I worked only in simulation with low-voltage stand-ins, as the safety notice asks. Nothing was connected to mains.

2. Design approach

The simulation uses an Arduino Uno in Wokwi. Two potentiometers stand in for the sensors (knob = 0-300 V on A0, 0-20 A on A1). A relay module switches the load, shown as a yellow LED. A 16x2 LCD, four buttons (MENU, UP, DOWN, RESET) and the serial port are the user interface. I kept the logic simple on purpose: the device is always in one of three states - **WAITING** (relay off, countdown), **ON**, or **TRIPPED** (relay off until RESET). The main loop runs every 20 ms: measure, decide, switch the relay and LEDs, read buttons and PC, refresh the LCD.

Choices I made and why: the relay is always off at power-up and waits 5 s before connecting; a short spike should not cut power, so every trip needs the problem to last a minimum time; very large faults trip immediately; over-current stays off until a person presses RESET (it may be a real short), while voltage faults reconnect automatically; reconnecting needs the voltage to be 5 V inside the limits, so the relay does not flutter near a limit.

3. Implementation

All code is one Arduino sketch (src/). Main parts: a protection function with the trip rules, a button/menu function, LCD drawing, and a serial command handler. Four settings (over-voltage, under-voltage, max current, reconnect wait) can be changed with the buttons or by commands: STATUS, GET, SET OV 240, SET UV/OC/WAIT, RESET, STREAM ON/OFF. The device also sends an EVENT line by itself whenever it trips or reconnects. Bad values (out of range, or over-voltage less than 20 V above under-voltage) are rejected.

4. Results

Situation	Behaviour (default settings)
Power-up	Relay off, 5 s countdown, then ON. Countdown stays at 5 s while voltage is outside the safe band (seen in Wokwi at 254.5 V).
Over-voltage	above 250 V for 20 ms: trip, reconnect after 5 s of good grid. 287.5 V or more: trip at once.
Under-voltage	below 180 V for 0.5 s: trip, then reconnect the same way.
Over-current	above 10 A for 2 s, or 30 A and more at once: trip and stay tripped until RESET (seen in Wokwi: LCD showed TRIP:OC RST after the current was lowered).
PC link	Settings change from the Serial Monitor take effect immediately; wrong values give an ERROR message.

Besides running it in Wokwi, the same logic was checked on a PC by an automated test with a fake Arduino and simulated knob values (timing, reconnect, latching, menu limits, serial commands); the test program was written with AI help. Screenshots and logs are in test_evidence/.

5. What is not done / limits

It is a simulation only: no real voltage or current sensor, no real relay, nothing measured on hardware. The knobs give an RMS value directly, so real AC sampling is not tested. Settings are lost when the simulation restarts (no EEPROM saving). A relay and a 20 ms loop cannot stop microsecond spikes - a real product also needs a fuse and a surge protector (MOV). There is no allowance for motor start-up current except the 2 s delay. The relay switch-off time was not measured and I have taken help of AI in different areas.